

Quantum Mechanics III

Lecture 8

Midterm Review

- Operational setting
- Mathematical representation
- Dynamics and measurement
- Consistency conditions
- Example

■ Operational setting

Preparation and measurement procedures are distinguished from their matrix representations.

$$P \rightarrow \rho \rightarrow \mathcal{E} \rightarrow \mathcal{M}_k \rightarrow p(k)$$

■ Mathematical representation

Positivity, trace, and tensor-product conventions are fixed before the main calculation.

$$CP : J_{\mathcal{E}} \geq 0$$

■ Dynamics and measurement

The map from a state to observed probabilities is written explicitly for the experiment under discussion.

$$\text{counts} \sim \text{Multinomial}(N, C_{\text{det}} p)$$

■ Consistency conditions

Complete positivity, normalization, Hermiticity, and appropriate classical limits are checked as needed.

■ Example

The example is worked for a qubit or two-mode system before extending the notation to higher dimension.

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Reading: Weeks 1–7 notes; Midterm formula sheet; QDC-26 summary